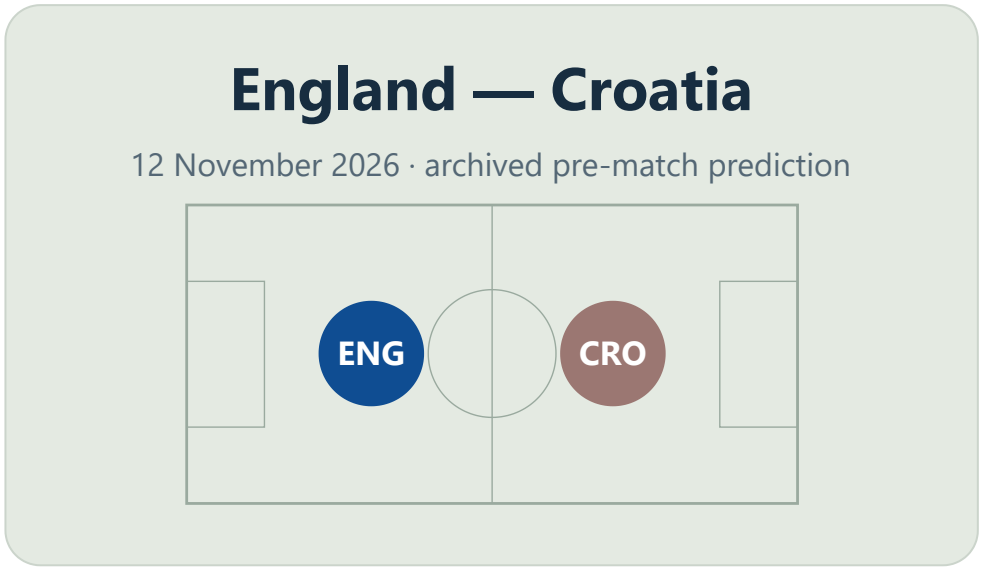
01 / A REAL FORECAST

The home team. The stronger team. The next result?

Can historical home-win patterns help us predict a match once we account for team strength?

Ruiyi Gao · Zhenglin Chen · Wenyu Jiang
Programming with Python · Fall 2026 · 睿信2526班



The fixture is real; the pitch is illustrative. Forecasts were sealed on 2 November 2026. Actual outcomes are not evaluated in this study.

One archive. Different questions, different samples.

Preserve the original records, resolve eligibility, then separate comparison from learning.

REGISTERED SNAPSHOT

49,945

matches, 1872–2026
Original files stay unchanged.

PREPARED THROUGH 2024

48,102

included records
Audit dates, scores, venues and duplicates.

IDENTITY-REVIEWED COMPARISON

24,917

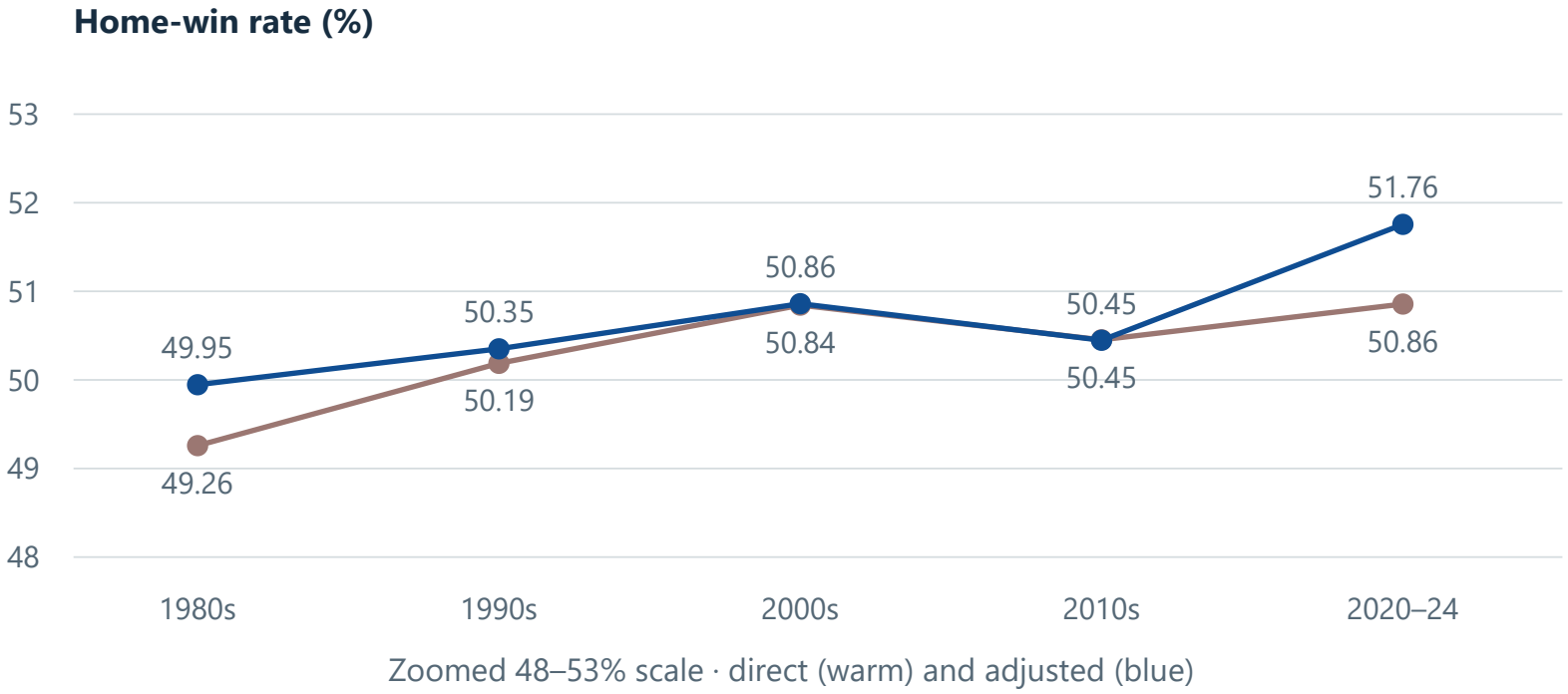
non-neutral matches, 1980–2024
Era, match-type and regional comparisons.

38,919 strength-learning matches from 1950, including earlier and neutral matches.

The comparison cohort and strength-learning set overlap; they are not successive exclusions or independent samples. The 117 project-lead-accepted records are retained, not presented as an independent official audit.

03 / DESCRIBE BEFORE EXPLAINING

A different home-win rate can mean a different mix of matches.



> Era rates are close in this archive. Their direction is unresolved under the tested long-term common-shock assumptions.

24,917 non-neutral matches, 1980–2024; the last era spans five years. These are descriptive estimates. Model-conditional checks do not establish a sustained era or match-type effect; regional contrasts depend on their stated reference and do not rank football ability.

04 / FROM RECORDS TO A PREDICTION

Predict first. Learn from the result afterwards.

Both opponents contribute information. A home result is interpreted in light of the venue and the opponent.

01 / BEFORE

Earlier matches

What was known before this date?

→

02 / ESTIMATE

Uncertain strength

Gradual change, possible jumps, both teams.

→

03 / PREDICT

Venue + context

Map the strength distribution to probabilities.

→

04 / OBSERVE

Actual result

All games on this date are predicted first.

→

05 / UPDATE

The next match

Learn from observed results, never predicted labels.

> Update both teams for later matches. Never use the model's own predicted result as if it had actually happened.

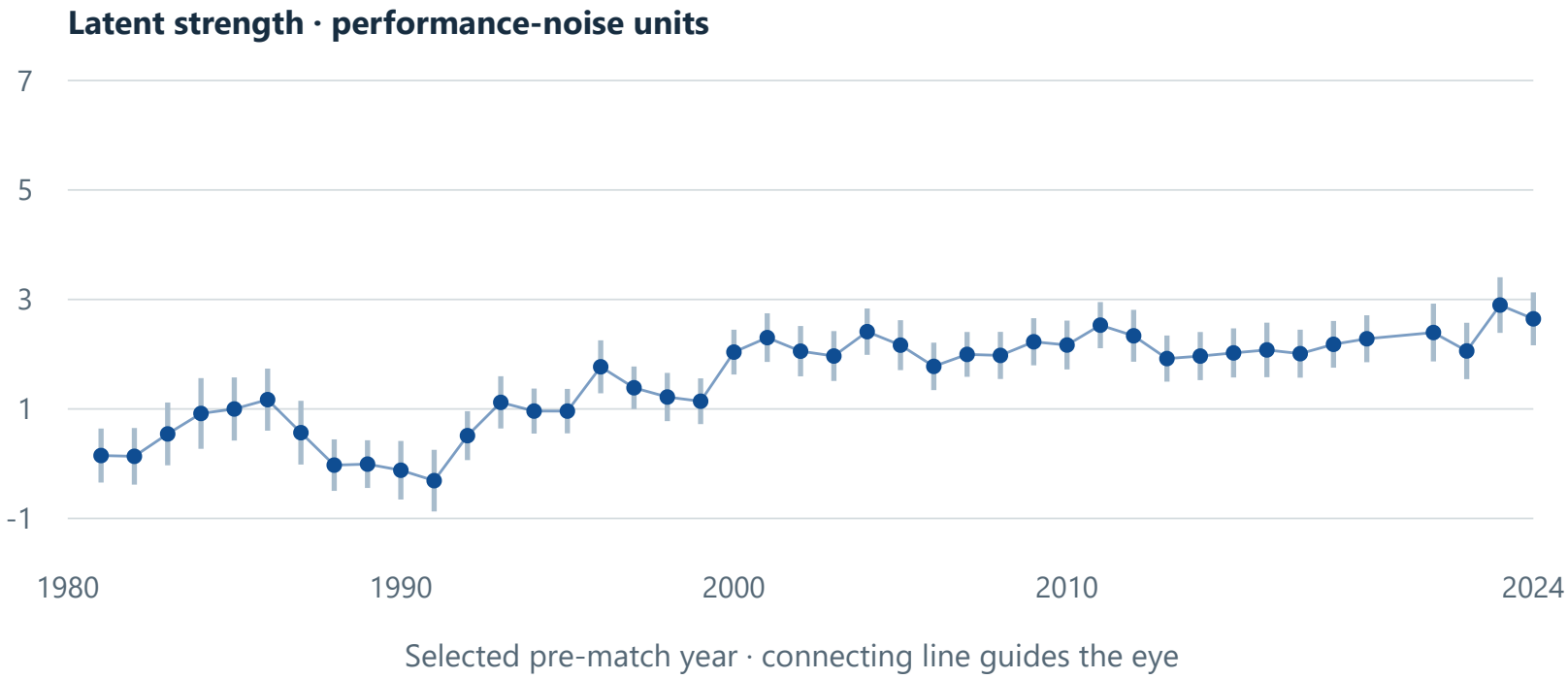
earlier results → uncertain strength → probability

The joint model receives a full strength-difference distribution. It estimates context separately; the rating model's home term is excluded from that input.

This is a workflow animation, not a live model refit. Friendly results carry less information; repeated surprising results can support a lasting change, without a guaranteed detection speed.

05 / THE MODEL'S OUTPUT

Strength is a changing estimate, not a permanent label.

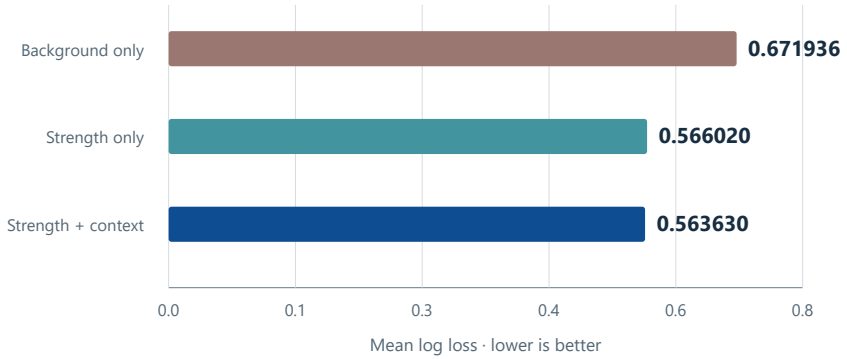


Japan · latest shown: 2024-11-19 · estimated strength 2.646 · marginal SD 0.483

1980–2024; latest eligible match per team/year, considering both roles. Marks are archived pre-match estimates, not true ability. Whiskers are marginal mean ±1 SD including jump variance, not confidence intervals. Lines only connect selected dates; gaps do not create observations.

06 / A FAIR COMPARISON

Strength adds more. Context adds a smaller increment.



JOINT VS BACKGROUND ONLY

16.12%

relative reduction in log loss

JOINT VS STRENGTH ONLY

0.42%

additional relative loss reduction

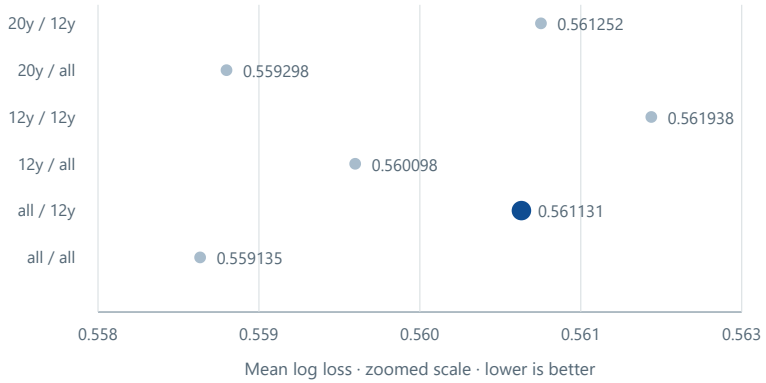
Different denominators: these are not additive shares of predictive information.

Joint binary accuracy: **70.01%**.

Same 21,413 matches, 1990–2024. Background-only / strength-only / joint use the same binary target: home win versus draw or away win. Historical data were reused during development. No real-data dependence-adjusted interval establishes persistence of the small gain.

07 / HOW MUCH HISTORY?

Keeping history worked best among the choices we tested.



Controlled cost comparison · rating history fixed

312.58 → 105.29 s

Use 12 years for outcome fitting.

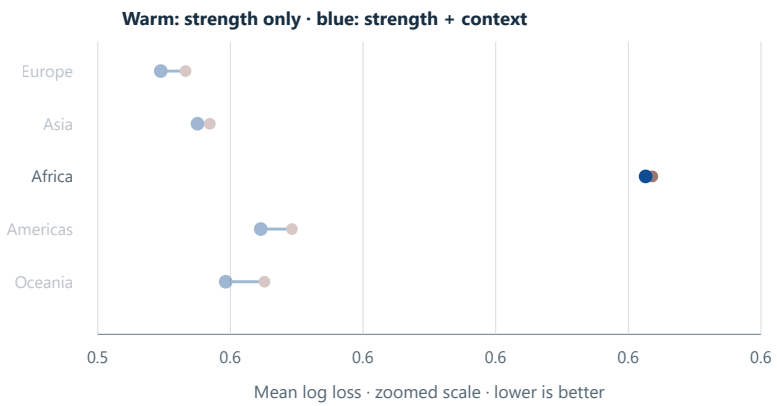
> **66.31% less fitting time**
> with a 0.3571% relative increase in loss.

This is one component's cost, not the total pipeline. It does not identify a universal best window.

Main comparison: 9,727 matches, 2010–2024. Dots show six discrete choices; loss scale is zoomed. Old results do not have equal current influence in a changing-strength model. Timings shown are the two controlled R10 runs; other batches are not treated as standardized benchmarks.

08 / WHERE DOES IT HOLD UP?

The pooled improvement does not tell the whole regional story.



AFRICA / HOME REGION

2,058 matches

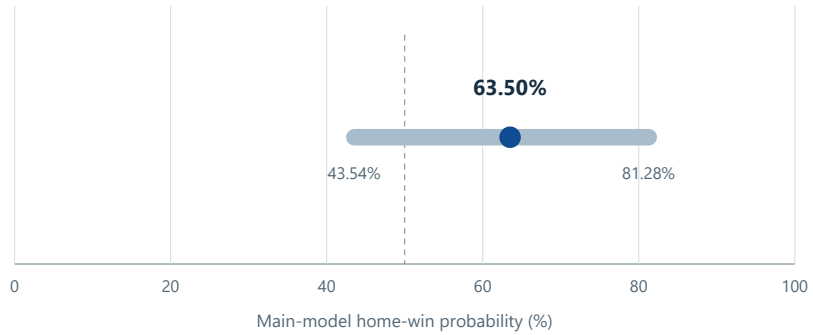
The context increment reversed after some deletions, including 2011 or matches involving Egypt. A small pooled bias can hide opposing probability-bin errors.

Joint bias: +0.6854 pp (predicted − observed)

Same main sample of 9,727 matches, 2010–2024, grouped by home region. Loss scale is zoomed; lower is better. Sensitivity checks held predictions fixed and changed evaluation composition, without refitting the learning history. These are not regional rankings, causal effects, or real-world confidence intervals.

09 / BACK TO THE NEXT MATCH

A forecast is a probability with a stated scope.



MAIN BINARY MODEL / 2026-11-12

63.50%

Home win; otherwise draw or away win.

The strength-induced range crosses 50%. A central pick can be fragile.

Sealed 2 November 2026; fixtures 12–17 November. Strength uses accepted results through 6 October 2026; outcome coefficients through 29 December 2025. 52 targets: 49 main binary estimates, three withheld for venue scope. Central 90% ranges concern latent strength with coefficients fixed; they exclude parameter/model error and do not guarantee accuracy. Actual outcomes remain unevaluated.

From historical patterns to a prediction we can inspect.

01 / UNDERSTAND THE DATA

Rates depend on who played.

Compare compositions before interpreting differences across time, match types or regions.

02 / LEARN FROM HISTORY

Strength carries the larger increment.

Track changing ability and uncertainty, then test what context adds on the same matches.

03 / STATE THE LIMITS

Probability is more informative than a confident pick.

Keep the issued forecasts unchanged. Next: evaluate new outcomes, improve calibration, and extend the joint target to win–draw–loss.

Traceable inputs. Comparable predictions. Explicit uncertainty.

The contribution is implementation, data/identity curation and common-sample empirical analysis, rather than invention of the cited rating principles. Static figure trials adapt Chen Liu's figures4papers (CC BY-NC 4.0); interactive SVG/JavaScript is implemented locally. Thank you.